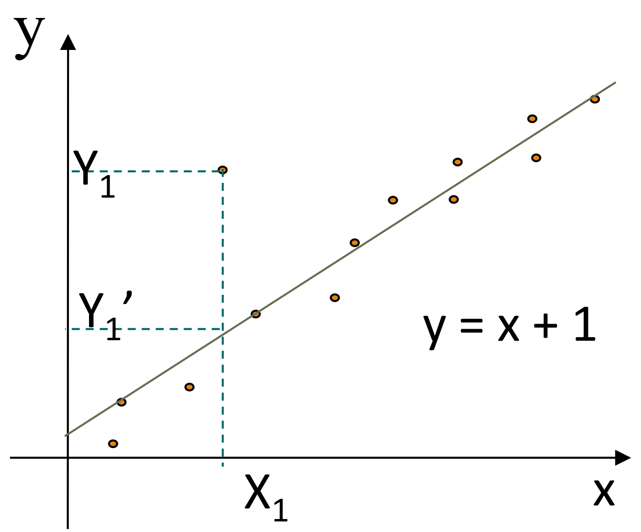
Parametric Data Reduction: Regression Analysis

- Regression analysis: A collective name for techniques for the modeling and analysis of numerical data consisting of values of a ***dependent variable*** (also called ***response variable*** or *measurement*) and of one or more *independent variables* (also known as ***explanatory variables*** or ***predictors***)



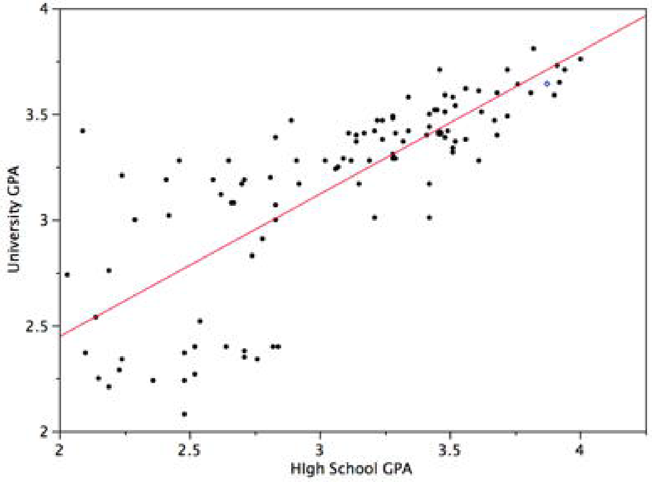
- The parameters are estimated to give a "**best fit**" of the data

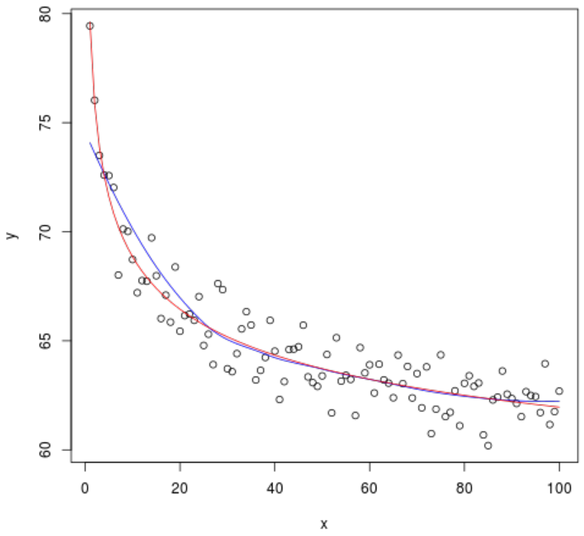
- Most commonly the best fit is evaluated by using the ***least squares method***, but other criteria have also been used

- Used for prediction (including forecasting of time-series data), inference, hypothesis testing, and modeling of causal relationships

Linear and Multiple Regression

- <u>Linear regression</u>: $Y = w\,X + b$
  - Data modeled to fit a straight line
  - Often uses the least-square method to fit the line
  - Two regression coefficients, *w* and *b*, specify the line and are to be estimated by using the data at hand
  - Using the least squares criterion to the known values of $Y_1, Y_2, \ldots, X_1, X_2, \ldots.$
- <u>Nonlinear regression</u>:
  - Data are modeled by a function which is a nonlinear combination of the model parameters and depends on one or more independent variables
  - The data are fitted by a method of successive approximations





Multiple Regression and Log-Linear Models

- <u>Multiple regression</u>: $Y = b_0 + b_1 X_1 + b_2 X_2$
  - Allows a response variable Y to be modeled as a linear function of multidimensional feature vector
  - Many nonlinear functions can be transformed into the above
- <u>Log-linear model</u>:
  - A math model that takes the form of a function whose logarithm is a linear combination of the parameters of the model, which makes it possible to apply (possibly multivariate) linear regression
  - Estimate the probability of each point (tuple) in a multidimension space for a set of discretized attributes, based on a smaller subset of dimensional combinations
  - Useful for dimensionality reduction and data smoothing